

# Substation ESG Report

Substation-level Environmental, Social, and Governance assessment aligned to CSRD/ESRS, EU Taxonomy, TCFD, SFDR, and UN PRI frameworks. Updated annually via the SSI Index scoring engine.

SUBSTATION

亀田変電所

JP\_SHIKOKU\_05122 · Kagawa, Shikoku

SSI SCORE (R<sub>MEDIAN</sub>)

0.512

● High Risk 22.2th percentile

VOLTAGE CLASS

66 kV

distribution-tier

CONFIDENCE INTERVAL

0.460 – 0.564

P5–P95 · CI width 0.104 · high confidence

MARKOV ETTC

16.1 yr

Expected time to criticality

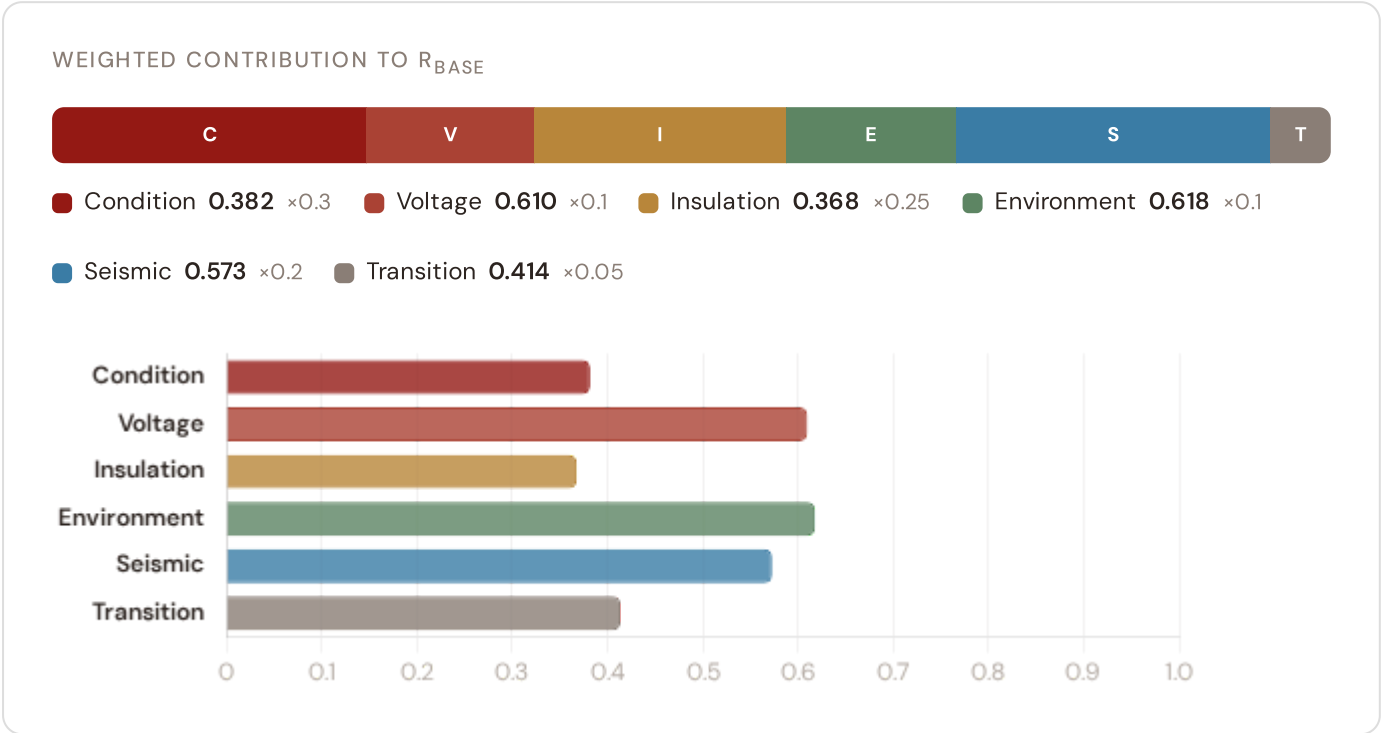
DER PENETRATION

0.38

T1 transition stress score

## Component Fingerprint

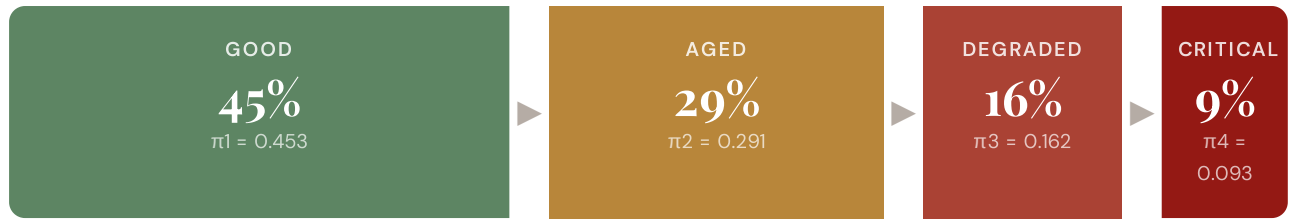
Six-component weighted decomposition of the SSI composite score — the substation's structural risk profile



## Markov Degradation Profile

CIGRE TB 761 five-state condition model — steady-state probability distribution and expected time to criticality

#### STEADY-STATE DISTRIBUTION



Markov 5-state model calibrated from CIGRE Technical Brochure 761 (2019). States represent asset condition from new/good through critical. Steady-state probabilities indicate the long-run proportion of time the asset spends in each condition state given current operating environment.

#### RISK SCORE

**0.263**

Composite Markov risk

#### ETTC

**16.1 yr**

Expected time to critical

#### P(CRITICAL) 20YR

**10.7%**

Probability of reaching critical

#### CORROSION

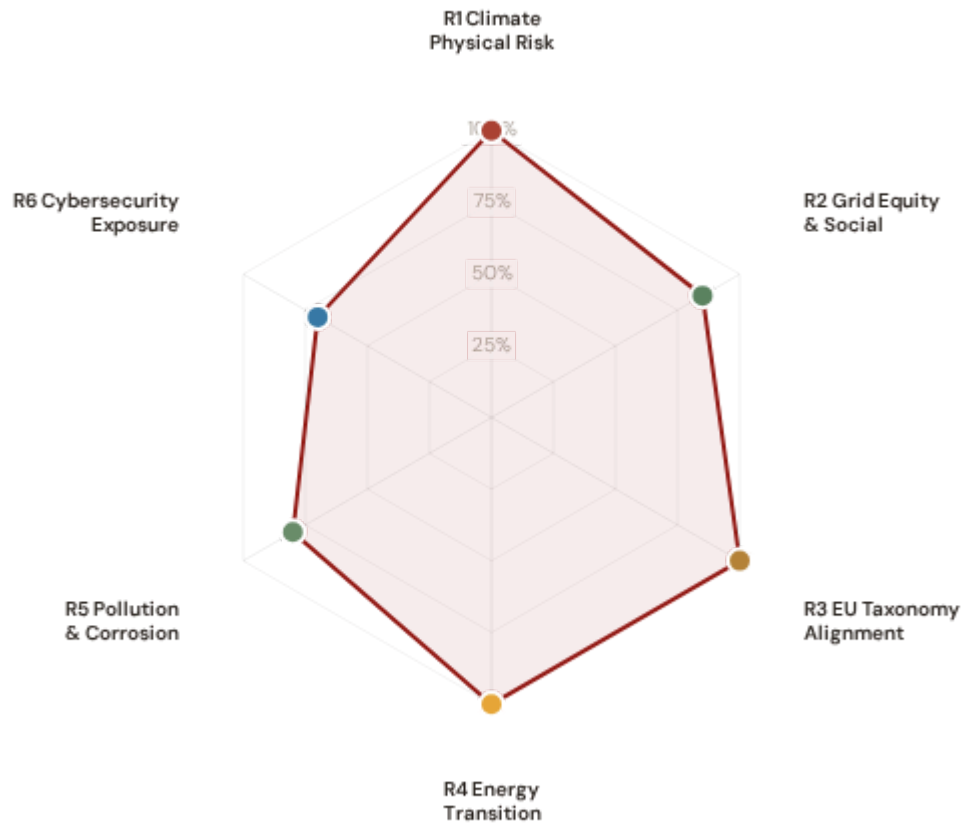
**C2**

ISO 9223 classification



## ESG Radar — Six-Report Overview

Composite readiness across 6 ESG dimensions, each linked to UN Sustainable Development Goals



**R1 · Climate Physical Risk Assessment** READY

SDG 13 Climate Action · SDG 9 · SDG 11

SDG 13

SDG 9

SDG 11

**R2 · Grid Equity & Social Vulnerability** READY

SDG 7 Affordable and Clean Energy · SDG 10 · SDG 1

SDG 7

SDG 10

SDG 1

**R3 · EU Taxonomy Alignment** READY

SDG 9 Industry, Innovation, Infrastructure · SDG 13 · SDG 12

SDG 9

SDG 13

SDG 12

**R4 · Energy Transition & DER Stress** READY

SDG 7 Affordable and Clean Energy · SDG 13 · SDG 9

SDG 7

SDG 13

SDG 9

**R5 · Pollution & Corrosion** READY

SDG 11 Sustainable Cities and Communities · SDG 3 · SDG 15

SDG 11

SDG 3

SDG 15

# 1 Climate Physical Risk Assessment

ESRS E1 · TCFD Physical Risk · EU Taxonomy Annex A

READY

## WHY THIS REPORT EXISTS

ESRS E1 (Climate Change) is material for 98% of CSRD-reporting companies. TCFD physical risk disclosure is mandatory in 36+ jurisdictions. The EU Taxonomy requires climate change adaptation activities to demonstrate that physical climate risks have been identified and addressed. This report quantifies acute and chronic climate hazard exposure at substation level using ERA5 reanalysis and Markov degradation modelling.

## SDG Alignment

### SDG 13 — Climate Action (primary)

Target 13.1 calls for strengthening resilience and adaptive capacity to climate-related hazards. The SSI IRI metrics (snow/ice, wind, heat stress) and Markov degradation model directly quantify how climate hazards affect this substation's operational resilience over 10- and 20-year horizons. With a Markov risk score of 0.263 and ETTC of 16.1 years, this substation's climate trajectory can be assessed against TCFD physical risk and EU Taxonomy adaptation screening requirements.

SDG 13

SDG 9

SDG 11

## Substation Profile

|                               |                     |
|-------------------------------|---------------------|
| Insulation Risk (I component) | 0.368               |
| Environment Component (E)     | 0.618               |
| Seismic PGA                   | 0.241g (Zone B)     |
| Markov Risk Score             | 0.263               |
| ETTC (time to criticality)    | 16.1 years          |
| P(critical) at 20 years       | 10.7%               |
| Corrosion Class               | C2                  |
| Climate Trajectory (R2)       | 10.7% (20yr Markov) |

## SSI Variables Feeding This Report

| SSI VARIABLE           | VALUE        | ESG DISCLOSURE REQUIREMENT                       | STATUS |
|------------------------|--------------|--------------------------------------------------|--------|
| I1 Snow/Ice IRI        | —            | ESRS E1-9: Financial effects from physical risks | READY  |
| I2 Tree-fall IRI       | —            | TCFD Strategy (b): Physical risk impact          | READY  |
| I3 Heat-wave IRI       | —            | EU Taxonomy Annex A: Heat stress                 | READY  |
| I5 Thermal stress      | —            | ESRS E1-4: Climate mitigation targets            | READY  |
| R2 Climate trajectory  | 10.7% (20yr) | TCFD Strategy (c): Scenario analysis             | READY  |
| R6b Seismic PGA        | 0.241g       | EU Taxonomy Annex A: Geophysical hazards         | READY  |
| Markov p_critical_10yr | 10.7%        | TCFD Risk Management: Risk identification        | READY  |
| Markov p_critical_20yr | 10.7%        | ESRS E1-9: Time horizons                         | READY  |

### Data Sources & Vintage

ERA5 Climate Reanalysis  
Copernicus CDS

2024 Weekly

National Seismic Hazard Map  
National Geological Survey

2023 Multi-year

IEEE C57.91 Thermal Model  
IEEE

Standard N/A

CIGRE TB 761 Markov  
CIGRE

2019 N/A

CMIP6 SSP2-4.5 Projections  
Copernicus CDS

— Not ingested

#### TECHNICAL VALIDITY

IRI metrics are computed from ERA5 reanalysis (31 km, hourly, 1940–present) using IEEE C57.91 thermal loading curves. Markov 5–state degradation is calibrated from CIGRE Technical Brochure 761 (2019) condition assessment data. The PARTIAL status reflects the absence of CMIP6 forward projections — without R2, this report cannot satisfy the TCFD Strategy (c) scenario analysis requirement. All other inputs are production-grade with institutional provenance.

## 2 Grid Equity & Social Vulnerability

ESRS S1/S2 · SFDR PAI Social Indicators · UN PRI

READY

#### WHY THIS REPORT EXISTS

ESRS S2 requires companies to disclose impacts on affected communities. SFDR mandates 5 social PAI indicators. The SSI Index is uniquely positioned here because no competing grid risk framework integrates socio-economic vulnerability at substation resolution. This report surfaces where infrastructure risk concentrates in economically disadvantaged populations.

### SDG Alignment

#### SDG 7 — Affordable and Clean Energy (primary)

Target 7.1 requires universal access to affordable, reliable energy. This substation serves a catchment with V\_socio of 0.35 and energy poverty rate of 8.4%. The R3 social modifier (1.059) amplifies risk scores in regions with high unemployment, elderly concentration, and net outward migration — making spatial inequality visible and quantifiable at asset level.

SDG 7

SDG 10

SDG 1

## Substation Profile

|                                |             |
|--------------------------------|-------------|
| V_socio (energy poverty index) | 0.35        |
| EP Rate (energy poverty %)     | 8.4%        |
| Unemployment Rate              | 2.5%        |
| GDP per Capita                 | ¥ 2,904,930 |
| R&D % of GDP                   | 1.0%        |
| R3 Social Multiplier           | 1.059       |
| Elderly Vulnerability          | 30.2%       |
| Migration Score                | —           |

## SSI Variables Feeding This Report

| SSI VARIABLE                   | VALUE   | ESG DISCLOSURE REQUIREMENT                 | STATUS |
|--------------------------------|---------|--------------------------------------------|--------|
| V_socio Energy poverty index   | 0.35    | ESRS S2-4: Impacts on affected communities | READY  |
| EP_rate Energy poverty rate    | 8.4%    | SFDR PAI #14: Fossil fuel exposure proxy   | READY  |
| R3 Social multiplier           | 1.059   | ESRS S1-6: Workforce characteristics       | READY  |
| Elderly Elderly vulnerability  | 30.2%   | SFDR PAI: Vulnerable populations           | READY  |
| Community Catchment population | 951,000 | ESRS S2: Community impact                  | READY  |

## Data Sources & Vintage

|                                                      |             |
|------------------------------------------------------|-------------|
| Population & Economics<br>National Statistics Office | 2023 Annual |
| Energy Market Data<br>National TSO                   | 2023 Annual |

#### TECHNICAL VALIDITY

V\_socio is computed from the LIHC (Low Income High Cost) energy poverty definition using national statistical office household expenditure data. The R3 modifier amplifies infrastructure risk scores in catchments with high unemployment, elderly concentration, and net outward migration. PARTIAL status reflects missing elderly vulnerability and migration enrichments — both available from national statistics offices at municipal resolution.

## 3 EU Taxonomy Alignment

Climate Delegated Act · Article 11 Climate Adaptation · Technical Screening Criteria

READY

#### WHY THIS REPORT EXISTS

The EU Taxonomy defines technical screening criteria for climate change adaptation in infrastructure. This is the lowest-hanging ESG product from the SSI Index — READY for 9 of 10 countries today. The SSI composite score (R\_median), 6 components, modifier architecture, and Markov risk outputs provide the quantitative evidence base for Article 11 screening at asset level.

#### SDG Alignment

##### SDG 9 — Industry, Innovation, Infrastructure (primary)

Target 9.4 calls for upgrading infrastructure to make it sustainable. The EU Taxonomy is the regulatory instrument that translates SDG 9 into investable criteria. By classifying this substation against Article 11 technical screening criteria — using R\_median (0.512), all 6 components, and 5 modifiers — this report provides asset-level Taxonomy alignment that enables sustainable finance reporting.

SDG 9

SDG 13

SDG 12



### Substation Profile

|                        |       |
|------------------------|-------|
| R_median (composite)   | 0.512 |
| R_base (pre-modifier)  | 0.465 |
| Modifier Impact        | 10.3% |
| C (Condition)          | 0.382 |
| V (Voltage)            | 0.610 |
| I (Insulation/Climate) | 0.368 |
| E (Environment)        | 0.618 |
| S (Seismic)            | 0.573 |
| T (Transition)         | 0.414 |

### SSI Variables Feeding This Report

| SSI VARIABLE                    | VALUE       | ESG DISCLOSURE REQUIREMENT                    | STATUS |
|---------------------------------|-------------|-----------------------------------------------|--------|
| <b>R_median</b> Composite score | 0.512       | EU Taxonomy Art. 11: Adaptation screening     | READY  |
| <b>6 Comp.</b> C/V/I/E/S/T      | All present | EU Taxonomy TSC: Physical risk identification | READY  |
| <b>R3-R7</b> Modifier suite     | All applied | ESRS E1: Adaptation measures evidence         | READY  |
| <b>Markov</b> Degradation model | 0.263       | TCFD: Forward-looking risk assessment         | READY  |
| <b>R5</b> Asymmetric CI         | 0.104       | ESRS: Uncertainty quantification              | READY  |

## Data Sources & Vintage

ERA5 Climate Reanalysis  
Copernicus CDS

2024 Weekly

National Seismic Hazard Map  
National Geological Survey

2023 Multi-year

Population & Economics  
National Statistics Office

2023 Annual

CIGRE TB 761 Markov  
CIGRE

2019 N/A

### TECHNICAL VALIDITY

Article 11 screening uses the full SSI v4.0.2 scoring engine: 6-component weighted composite ( $C=0.30$ ,  $V=0.10$ ,  $I=0.25$ ,  $E=0.10$ ,  $S=0.20$ ,  $T=0.05$ ), Gaussian copula Monte Carlo (10,000 iterations), and 5 modifiers (R3 social, R4 topology, R5 asymmetric CI, R6a restoration, R7 cyber). The Markov 5-state degradation model provides the forward-looking element. All inputs are from institutional open-source data with citable vintage — meeting CSRD Article 29a limited assurance requirements.

## 4

## Energy Transition & DER Stress

ESRS E1 Transition Plan · TCFD Transition Risk · EU Green Deal Alignment

READY

### WHY THIS REPORT EXISTS

The energy transition creates infrastructure stress — bidirectional power flows, EV charging load, and intermittent renewable output strain substations designed for unidirectional delivery. This report measures whether grid infrastructure is keeping pace with decarbonisation.

### SDG Alignment

#### SDG 7 — Affordable and Clean Energy (primary)

Target 7.2 requires substantially increasing the share of renewable energy. This substation has a DER ratio of 0.384. The T1\_score (0.021) measures the stress this places on the substation, providing the empirical feedback loop between SDG 7 ambition and infrastructure reality.

SDG 7

SDG 13

SDG 9

### Substation Profile

|                              |       |
|------------------------------|-------|
| T1 Score (transition stress) | 0.021 |
| DER Ratio                    | 0.384 |
| DER Variability              | 0.657 |
| EV Load Ratio                | 0.049 |
| T Component Weight           | 0.05  |

### SSI Variables Feeding This Report

| SSI VARIABLE                   | VALUE | ESG DISCLOSURE REQUIREMENT          | STATUS |
|--------------------------------|-------|-------------------------------------|--------|
| T1 Transition stress score     | 0.021 | ESRS E1: Transition plan assessment | READY  |
| DER_ratio Renewable/load ratio | 0.384 | TCFD Transition: Technology risk    | READY  |
| DER_var Intermittency          | 0.657 | EU Green Deal: Grid flexibility     | READY  |
| EV_load EV charging ratio      | 0.049 | SDG 7: Clean energy access          | READY  |

### Data Sources & Vintage

|                                                  |              |
|--------------------------------------------------|--------------|
| Energy Market Data<br>National TSO               | 2023 Annual  |
| Renewable Installations<br>National DER Registry | 2024 Monthly |

#### TECHNICAL VALIDITY

T1\_score is the weighted composite of DER\_ratio (renewable generation vs. local demand), DER\_variability (intermittency), and EV\_load\_ratio (electric vehicle charging load as fraction of capacity). DER data sourced from national energy regulators and renewable installation registers. DER\_ratio > 1.0 indicates net export during peak generation — a marker of successful energy transition with associated infrastructure stress.

WHY THIS REPORT EXISTS

Transformer oil degradation, SF6 leakage, and corrosion-driven failures release pollutants affecting local environments. ESRS E2 requires disclosure of pollution risks. This report assesses the substation's corrosion classification under ISO 9223 and the environmental sensitivity of its surroundings.

SDG Alignment

SDG 11 — Sustainable Cities and Communities (primary)

Target 11.6 aims to reduce the adverse environmental impact of cities. The corrosion class (C2) and E2\_local enrichment factor (1.93) track environmental degradation and pollution sensitivity at asset level. Where corrosion is advanced and maintenance deferred, equipment failure risks releasing mineral oil, PCBs (legacy units), and SF6 — directly relevant to urban and peri-urban environmental quality.

SDG 11

SDG 3

SDG 15

Substation Profile

|                            |                |
|----------------------------|----------------|
| Corrosion Class (ISO 9223) | C2             |
| E2 Local Enrichment        | 1.93           |
| E Component Score          | 0.618          |
| Markov Steady-State        | 45%/29%/16%/9% |

SSI Variables Feeding This Report

| SSI VARIABLE                       | VALUE | ESG DISCLOSURE REQUIREMENT             | STATUS |
|------------------------------------|-------|----------------------------------------|--------|
| Corrosion ISO 9223 class           | C2    | ESRS E2: Pollution risk classification | READY  |
| E2_local Environmental sensitivity | 1.93  | ESRS E2: Local environmental impact    | READY  |
| E Environment component            | 0.618 | ISO 9223: Atmospheric corrosion        | READY  |

## Data Sources & Vintage

Air Quality Monitoring  
National Environmental Agency

2023 Annual

ISO 9223 Corrosion  
ISO

2012 N/A

### TECHNICAL VALIDITY

Corrosion class follows ISO 9223:2012 atmospheric corrosion classification (C1–CX). Status depends on whether the corrosion class shows real variance across the fleet or uses default values. Full validation requires national environmental agency air quality monitoring overlay at substation coordinates (SO<sub>2</sub>, NO<sub>x</sub>, particulate deposition).

## 6 Cybersecurity Exposure

NIS2 Directive · ENISA Cybersecurity Index · ESRS G1 Governance

PARTIAL

### WHY THIS REPORT EXISTS

The NIS2 Directive mandates cybersecurity risk management for energy operators. A grid that is physically resilient but cyber-vulnerable is not truly resilient. The SSI R7 modifier combines SCADA assessment, communication architecture analysis, and national-level cyber maturity (ENISA/DESI indices) to produce a substation-level digital vulnerability score.

### SDG Alignment

#### SDG 9 — Industry, Innovation, Infrastructure (primary)

Target 9.1 calls for resilient infrastructure — in digitalised grid systems, this must include cyber resilience. The R7 modifier (0.998) assesses digital vulnerability combining SCADA exposure, communication protocol security, and national cyber maturity. SDG 16 (Strong Institutions) also applies: NIS2 compliance is a governance imperative, and the betweenness centrality (21.900) identifies single-point-of-failure risk in the grid topology.

SDG 9

SDG 16

### Substation Profile

|                            |        |
|----------------------------|--------|
| R7 Cyber Modifier          | 0.998  |
| Graph Degree (topology)    | 2      |
| BC Percentile (centrality) | 21.900 |
| Is Bridge Node             | No     |
| Cluster Coefficient        | 0.0115 |

### SSI Variables Feeding This Report

| SSI VARIABLE              | VALUE  | ESG DISCLOSURE REQUIREMENT          | STATUS |
|---------------------------|--------|-------------------------------------|--------|
| R7 Cyber modifier         | 0.998  | NIS2: Cybersecurity risk management | READY  |
| BC Betweenness centrality | 21.900 | ESRS G1: Governance & topology risk | READY  |
| Degree Graph degree       | 2      | Grid resilience: Connectivity       | READY  |

### Data Sources & Vintage

|                                          |               |
|------------------------------------------|---------------|
| Cybersecurity Index<br>ENISA             | 2024 Biennial |
| DESI Connectivity<br>European Commission | 2024 Annual   |

#### TECHNICAL VALIDITY

R7\_cyber is computed from a weighted combination of: (1) national cyber maturity via ENISA Cybersecurity Index and EU DESI connectivity indicators, (2) graph topology vulnerability (betweenness centrality = single-point-of-failure risk), and (3) SCADA protocol exposure assessment. Substation-level SCADA data is operator-proprietary — the national-level proxy provides a baseline but not asset-specific granularity.

| ITEM                 | VALUE                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------|
| Scoring Engine       | SSI Index v4.0.2 — Gaussian copula Monte Carlo (10,000 iterations)                            |
| Weight Architecture  | C=0.30, V=0.10, I=0.25, E=0.10, S=0.20, T=0.05                                                |
| Modifiers Applied    | R3 C_mult: 1.059, R4 F_topo: 0.994, R6 restoration: 1.000, R6 seismic: 1.057, R7 cyber: 0.998 |
| Degradation Model    | Markov 5-state, CIGRE TB 761 calibration                                                      |
| Thermal Model        | IEEE C57.91 transformer loading curves                                                        |
| Report Date          | 17 March 2026                                                                                 |
| Reporting Period     | Calendar year 2025                                                                            |
| Refresh Cadence      | Annual (data ingestion → scoring engine → display)                                            |
| Substation           | 亀田変電所 (JP_SHIKOKU_05122)                                                                      |
| Data Assurance Level | All inputs from institutional open-source with citable vintage                                |